

Emergent altermagnetic correlations in the checkerboard Hubbard model

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We study magnetic correlations in the half-filled Hubbard model on the checkerboard lattice by constrained-path quantum Monte Carlo. The hopping is spin independent, so the momentum distributions of the two spin species coincide at $U = 0$ and any splitting must be generated by the interaction. The two magnetic sublattices are exchanged by a fourfold rotation and by no translation, which is the symmetry condition for a compensated spin splitting. We find such a splitting above a finite interaction strength, with a momentum-space polarization that is odd under C_4 and even under the diagonal mirror at all 126 interacting parameter sets studied, identifying the d_{xy} channel. A control at vanishing anisotropy, where the model reduces to the square lattice and the splitting is forbidden, fixes the statistical floor of the estimator at $6.3(2) \times 10^{-4}$ per site, between 23 and 114 times below the measured signal. Finite-size analysis on lattices from 8×8 to 16×16 sites shows the equal-time structure factor falling as $1/N$ and the extrapolated staggered moment dropping below the resolution of the calculation, so the correlations remain short ranged on the lattices reached here.

I. INTRODUCTION

Altermagnets combine a vanishing net moment with a spin-split band structure, a combination long assumed to require either ferromagnetism or spin-orbit coupling [1, 2]. The splitting follows from the crystal symmetry: the two opposite-spin sublattices are connected by a rotation rather than by a translation or an inversion, so the spin degeneracy that a conventional antiferromagnet enforces is absent.

Most microscopic models of this state build the splitting into the Hamiltonian, usually through a spin-dependent hopping. The splitting is then present already at zero interaction and the question of whether interactions alone can produce it is left open. Two recent studies of the Lieb lattice state the gap directly, noting that unbiased studies establishing an altermagnet as the ground state of an interacting Hamiltonian are lacking [3, 4]. Both use methods with a controlled range of validity, unrestricted Hartree-Fock with exact diagonalization on small clusters in the first case and a functional renormalization group treatment of the weak-coupling limit in the second.

The checkerboard lattice offers a route to the same physics without putting it in by hand. It is a square lattice with diagonal bonds on alternating plaquettes, so the two sublattices are exchanged by a fourfold rotation about a plaquette center but by no translation. The hopping is spin independent, and the interaction is a uniform on-site Hubbard U . Any spin splitting therefore appears only once magnetic correlations develop, which is the emergent route rather than the built-in one.

We report constrained-path quantum Monte Carlo (CPQMC) calculations for this model at half filling on lattices of 8×8 up to 16×16 sites. Three results follow. The momentum-resolved spin polarization is nonzero above a finite U and has d_{xy} symmetry at every interacting parameter set we examined. A null calculation at vanishing anisotropy, where symmetry forbids

the signal, places the statistical floor two orders of magnitude below the measured values. Finite-size analysis to 16×16 sites shows that the correlations stay short ranged on the lattices reached, so we describe them as correlations rather than as order.

II. MODEL AND METHOD

We consider

$$H = \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1)$$

with nearest-neighbor hopping $t_{ij} = t_0$ and diagonal hopping $t_1 + t_2$ along $(1, 1)$ and $t_1 - t_2$ along $(1, -1)$ on one sublattice, the two exchanged on the other. Energies are in units of $|t_0|$, and we set $t_0 = -1$ and $t_1 = 0.3$ throughout. The anisotropy is parametrized by δ through $t_2 = -\delta$, so that $\delta = 0$ restores the plain square lattice with uniform second-neighbor hopping and both sublattices become equivalent. All calculations are at half filling.

The hopping in Eq. (1) is spin independent by construction, so $n_\uparrow(\mathbf{k}) = n_\downarrow(\mathbf{k})$ at $U = 0$ for every δ .

We use CPQMC with a Trotter step $\Delta\tau = 0.01$, a projection length $\beta = 32$, and $N_w = 1000$ walkers. The constrained-path approximation introduces a bias controlled by the trial wave function, so we do not describe these results as exact. Doubling the projection length from $\beta = 32$ to $\beta = 48$ changes the order parameter by between 0.1% and 4.1% with no systematic drift, which we take as convergence in β .

The order parameter is the momentum-space spin polarization

$$\Delta_{\text{tot}} = \sum_{\mathbf{k}} |n_\uparrow(\mathbf{k}) - n_\downarrow(\mathbf{k})|, \quad (2)$$

following Ref. [5]. The absolute value inside the sum makes Δ_{tot} sensitive to a compensated splitting, for

which the unsigned sum $\sum_{\mathbf{k}} [n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})]$ vanishes. We quote Δ_{tot}/N so that different lattice sizes can be compared, and we note that Eq. (2) as written in Ref. [5] is a bare sum.

Because H commutes with the total spin, the polarization must be induced by a symmetry-breaking field. We add a staggered field h that couples with opposite sign to the two sublattices, which makes the trial state symmetry broken, and we report the behavior as a function of h . The constrained path is defined by the trial state, so this field biases the walkers toward the broken state, and the control described below is what establishes that the measured signal is not an artifact of that bias.

To classify the symmetry of $\Delta n(\mathbf{k}) = n_{\uparrow}(\mathbf{k}) - n_{\downarrow}(\mathbf{k})$ we use two dimensionless ratios,

$$A_{\text{odd}} = \frac{\sum_{\mathbf{k}} |\Delta n(\mathbf{k}) + \Delta n(R_{90}\mathbf{k})|}{\sum_{\mathbf{k}} |\Delta n(\mathbf{k})|}, \quad (3)$$

$$M_{\text{odd}} = \frac{\sum_{\mathbf{k}} |\Delta n(\mathbf{k}) + \Delta n(M\mathbf{k})|}{\sum_{\mathbf{k}} |\Delta n(\mathbf{k})|}, \quad (4)$$

where R_{90} is the fourfold rotation and M the diagonal mirror $(k_x, k_y) \rightarrow (k_y, k_x)$. A state odd under C_4 gives $A_{\text{odd}} = 0$ and even gives $A_{\text{odd}} = 2$. A $d_{x^2-y^2}$ state gives $M_{\text{odd}} = 0$ and a d_{xy} state gives $M_{\text{odd}} = 2$. These ratios use only the symmetry of the data, so unlike a projection onto a single lattice harmonic they are not sensitive to how the weight is distributed among higher harmonics of the same symmetry.

III. EMERGENT ALTERMAGNETIC CORRELATIONS

Δ_{tot} vanishes identically at $U = 0$ and becomes finite for $U > 0$ at every anisotropy studied, reaching Δ_{tot}/N between 2.2×10^{-3} and 9.2×10^{-2} over the interacting parameter sets. The splitting is compensated, so the state carries momentum-resolved spin polarization with no net moment.

The symmetry classification is unambiguous. Across all 126 interacting parameter sets at $L = 8, 10$, and 12 we obtain A_{odd} between 0.004 and 0.263 and M_{odd} between 1.24 and 2.00 . Restricted to $L = 12$ the ranges are 0.008 to 0.166 and 1.54 to 2.00 . The polarization is therefore odd under C_4 and even under the diagonal mirror at every parameter set, which is the d_{xy} channel. No parameter set is classified as $d_{x^2-y^2}$.

IV. NULL CONTROL AT VANISHING ANISOTROPY

Equation (2) sums absolute values, so statistical noise contributes with a definite sign and Δ_{tot} has a positive floor that does not vanish when the physics requires it to. Establishing the scale of that floor requires a calculation in which the signal is forbidden but the sampling is unchanged.

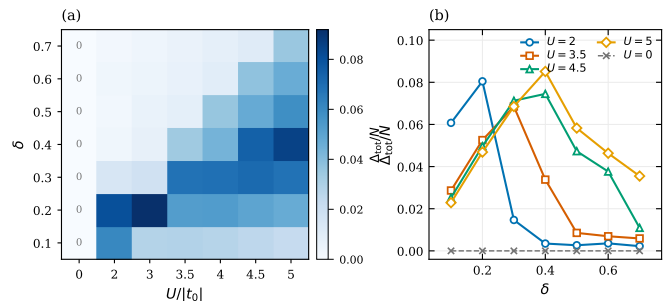


FIG. 1. Momentum-space spin polarization Δ_{tot}/N over the (U, δ) plane at half filling, $L = 12$. The $U = 0$ column is identically zero at every anisotropy, which is required because the hopping in Eq. (1) is spin independent. The signal appears only once the interaction is turned on, which is the emergent route rather than the built-in one.

The $U = 0$ column does not serve this purpose. There the Green function is obtained without auxiliary fields, so the zero is a matter of arithmetic rather than of statistics and it tests nothing about the sampling.

At $\delta = 0$ both diagonal bonds are equal on every site, the two sublattices become equivalent, and the model reduces to the square lattice with uniform second-neighbor hopping. Its Néel sublattices are related by a translation, so $n_{\uparrow}(\mathbf{k}) = n_{\downarrow}(\mathbf{k})$ identically and Δ_{tot} must vanish by symmetry at any U . Whatever is measured there is the statistical floor at production settings.

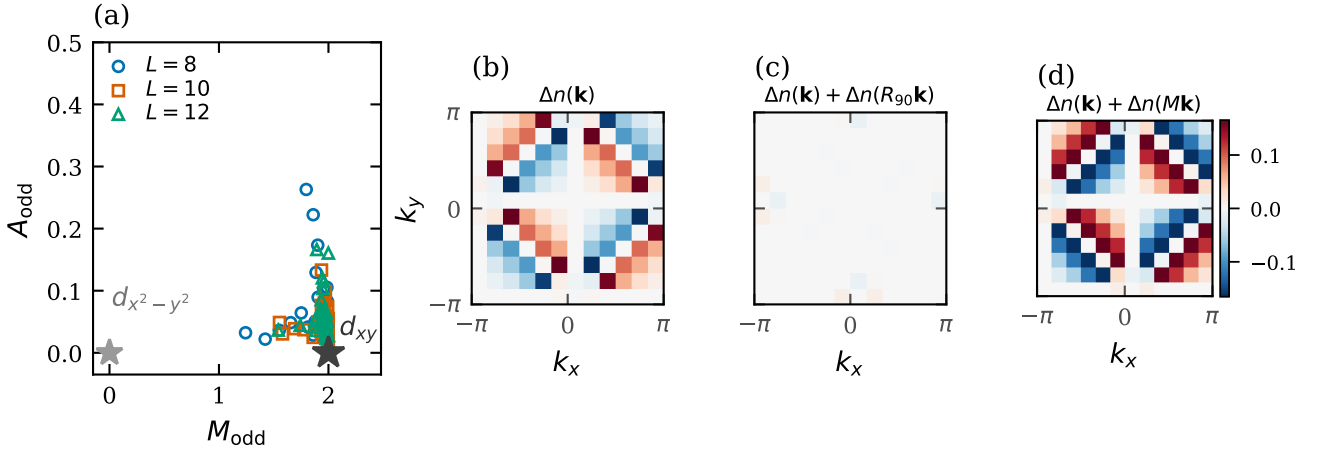
We find $\Delta_{\text{tot}}/N = 6.3(2) \times 10^{-4}$, averaged over $U = 2, 3, 3.5, 4, 4.5$, and 5 at $L = 12$. At $\delta = 0.3$ the measured signal ranges from 1.47×10^{-2} at $U = 2$ to 7.13×10^{-2} at $U = 4.5$, which is between 23 and 114 times this floor.

V. FINITE-SIZE BEHAVIOR OF THE MAGNETIC CORRELATIONS

The polarization of Sec. III is measured on finite lattices, and a nonzero value there does not by itself establish order in the thermodynamic limit. We therefore examine two quantities that distinguish a genuine order parameter from a finite-size remnant.

The first is the equal-time spin structure factor at the ordering wave vector, $S(\pi, \pi)/N$. Across $L = 8, 10$, and 12 and for densities from $n = 0.5$ to $n = 1$ at U up to 8 , the ratio between successive sizes falls between 0.39 and 0.54 , against 0.44 for a pure $1/N$ decay. A structure factor that scales as $1/N$ carries no long-range order, so this alone leaves the correlations short ranged on the lattices reached here.

The second is the extrapolation of the polarization to vanishing field. The $h \rightarrow 0$ intercept of Δ_{tot}/N is nonzero at each individual size and decreases monotonically with L , taking the values $0.03269, 0.02363, 0.01845$, and 0.01714 at $L = 8, 10, 12$, and 14 . On those four points alone the extrapolation is ambiguous. A fit in $1/L^2$ gives $\chi^2/\text{dof} = 2.1$ against 7.6 for a fit in $1/L$,



$$L = 12, U = 4, \delta = 0.4$$

FIG. 2. Symmetry classification of $\Delta n(\mathbf{k})$ for all 126 interacting parameter sets at $L = 8, 10$ and 12 . The ratios A_{odd} of Eq. (3) and M_{odd} of Eq. (4) place every cell at $A_{\text{odd}} \simeq 0$ and $M_{\text{odd}} \simeq 2$, which is odd under C_4 and even under the diagonal mirror and identifies the d_{xy} channel. No cell is classified as $d_{x^2-y^2}$.

half filling. $L = 14$ forms a curve only for $\delta \leq 0.3$; its single $\delta = 0.4$ cell is omitted.

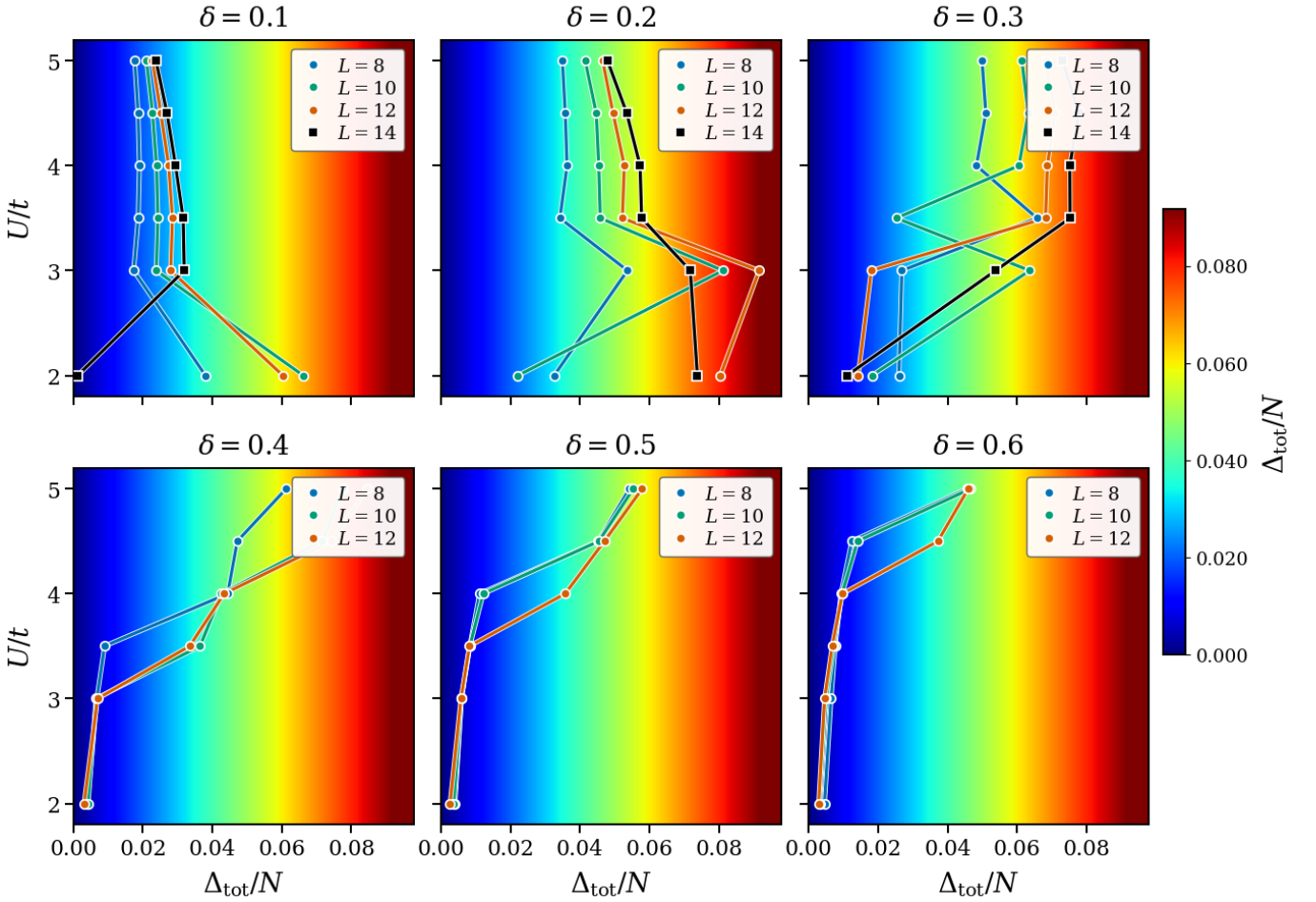


FIG. 3. Δ_{tot}/N against U at six anisotropies from $\delta = 0.1$ to 0.6 , for $L = 8, 10, 12$ and 14 . The background shades the value of Δ_{tot}/N on the same scale as the heat maps so that the two presentations can be read against each other. $L = 14$ forms a curve only for $\delta \leq 0.3$.

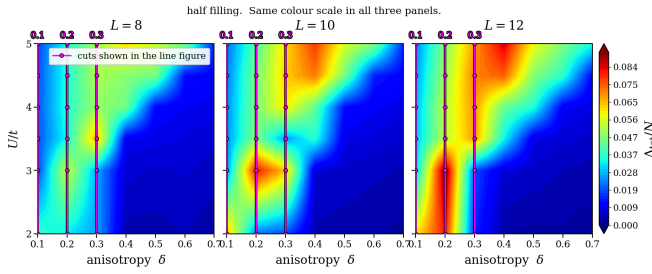


FIG. 4. Δ_{tot}/N over the (U, δ) plane with the cuts used in the preceding figure drawn on the map, $L = 12$.

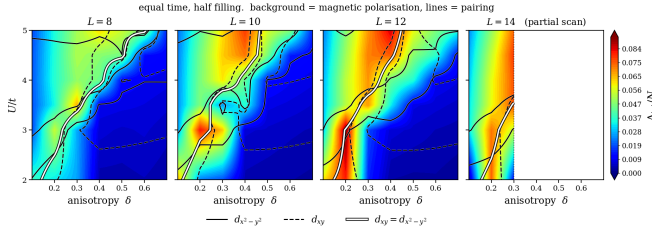


FIG. 5. The four lattice sizes overlaid, so that the size dependence of Δ_{tot}/N can be read at fixed U and δ .

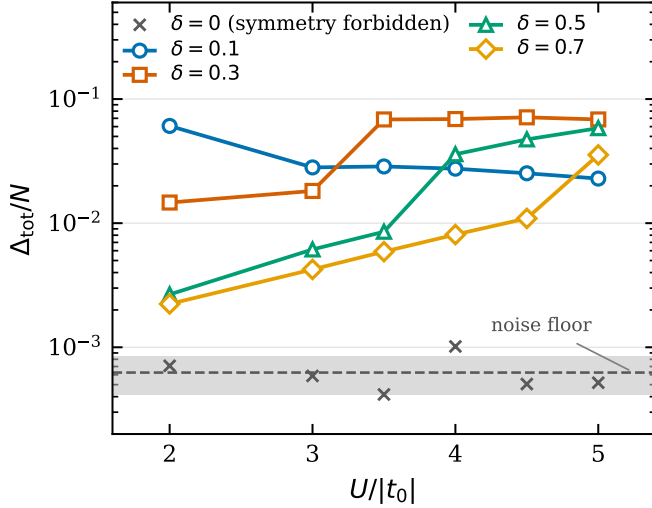


FIG. 6. Null control at vanishing anisotropy, logarithmic axis. At $\delta = 0$ the two sublattices are related by a translation and Δ_{tot} must vanish by symmetry at any U , so what is measured there is the statistical floor of the estimator at production settings. The floor is $6.3(2) \times 10^{-4}$ per site and the measured signal at $\delta = 0.3$ sits between 23 and 114 times above it.

which favors a finite intercept and therefore order.

The $L = 16$ calculation resolves the ambiguity. The two fits predict 0.01325 in $1/L$ and 0.01457 in $1/L^2$, and the measured value is 0.00734 ± 0.00109 . The drop from $L = 14$ is a factor of 0.43 where a $1/L$ law predicts 0.88, and including the point raises the $1/L^2$ χ^2/dof from 2.1 to 11.9. The intercept is a finite-size effect and the magnetic correlations remain short ranged.

This is a statement about the correlation length, not about the symmetry of the polarization. The classification of Sec. III rests on ratios of sums over the Brillouin zone at a fixed lattice size, so it is unaffected by the scaling of the correlation length.

VI. ROBUSTNESS AND COMPARISON

At half filling with periodic boundaries the noninteracting gap of Eq. (1) is zero at every L and δ we examined, so the occupied set is fixed by the trial state rather than by the spectrum. We repeated the calculation with antiperiodic boundaries, which shift the momentum mesh off the high-symmetry points and open a gap, at otherwise identical settings.

The sign of $\partial\Delta_{\text{tot}}/\partial U$ agrees between the two boundary conditions at $\delta = 0.1$, where both are negative, and at $\delta = 0.5$, where both are positive, with magnitudes matching to within a few percent despite the different meshes. At $\delta = 0.3$ the two disagree, and the antiperiodic value is consistent with zero. We read $\delta \approx 0.3$ as the crossover where the slope passes through zero, at which point the sign is most sensitive to the mesh, and we therefore do not quote a slope at that anisotropy.

VII. CONCLUSION

The half-filled checkerboard Hubbard model supports compensated spin-split correlations that are absent at $U = 0$ and are d_{xy} in symmetry at every parameter set examined. A control at vanishing anisotropy, where the splitting is forbidden by symmetry, places the statistical floor two orders of magnitude below the signal, which establishes that the measured polarization is not an artifact of the estimator or of the symmetry-broken trial state.

The mechanism uses no spin-dependent hopping and no spin-orbit coupling. It requires only a lattice whose magnetic sublattices are related by a rotation and a uniform Hubbard repulsion. The correlations remain short ranged on the lattices reached here, so we do not claim long-range order, and the constrained-path bias remains.

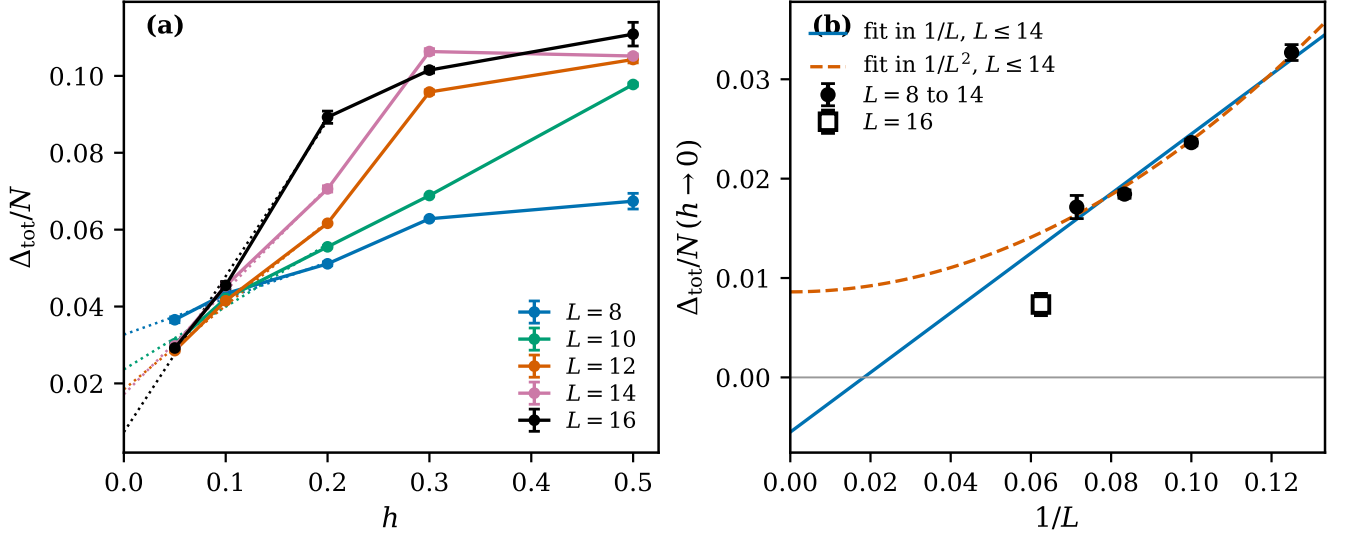


FIG. 7. (a) Δ_{tot}/N against the symmetry-breaking field h for $L = 8$ to 16 at $U = 4$ and $\delta = 0.3$, with the straight-line fits over the three smallest fields that define the $h \rightarrow 0$ intercept. (b) those intercepts against $1/L$. The two curves are fits to the $L \leq 14$ points alone, linear in $1/L$ and in $1/L^2$, extended to $L = 16$. They predict 0.01325 and 0.01457 there, and the measured value is 0.00733 ± 0.00106 . The drop from $L = 14$ is a factor of 0.43 where a $1/L$ law gives 0.88 , so the nonzero intercept seen at each individual size is a finite-size effect.

half filling, $L = 8$ to 14 . The $\delta = 0.1$ peak is gone at $L = 14$.

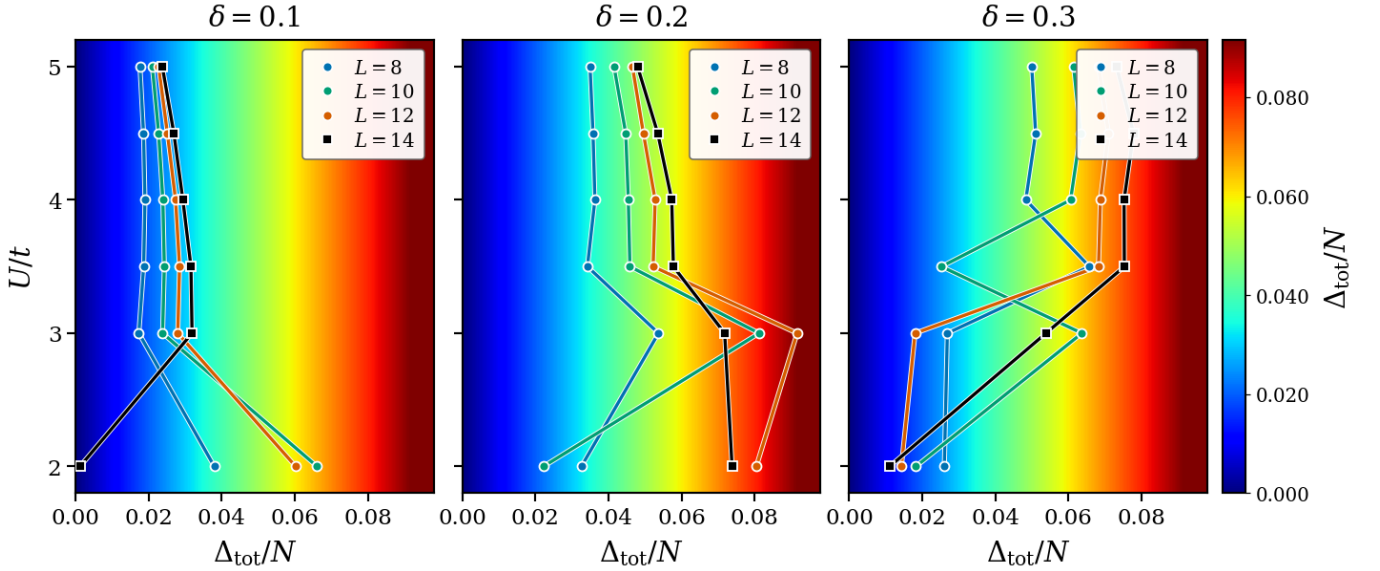


FIG. 8. Behavior at small U for $\delta = 0.1, 0.2$ and 0.3 across $L = 8$ to $L = 14$. The feature near $U = 2$ does not survive to $L = 14$ at $\delta = 0.1$, where Δ_{tot}/N falls to about twice the $\delta = 0$ floor and the correlation with $\sin k_x \sin k_y$ is lost. At $\delta = 0.3$ the decay with size is steady. The $\delta = 0.2$ column persists at every size.

Appendix A: Trial wave function and the selection of the constrained path

The constrained-path approximation replaces the sign problem with a bias fixed by the trial wave function, so the trial is part of the method rather than an implementation detail. We take the trial from an unrestricted

Hartree-Fock solution of Eq. (1) at the same U , δ , and filling, and we select among the solutions reached from different starting densities by the variational energy

$$E_{\text{HF}} = \text{Tr}(K\rho_{\uparrow}) + \text{Tr}(K\rho_{\downarrow}) + U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle, \quad (\text{A1})$$

with $\rho_\sigma = V_\sigma V_\sigma^\dagger$ built from the occupied columns of the mean-field eigenvectors and K the one-body matrix of Eq. (1).

Equation (A1) is not interchangeable with the sum of mean-field eigenvalues. Each eigenvalue already contains the mean field of the opposite spin, so that sum counts the interaction twice and orders competing solutions differently from the energy. The two criteria can select different states, and because the constrained path is fixed by whichever state is chosen, the selection propagates into every measured quantity.

The size of the effect is not small. Selecting by the eigenvalue sum instead of Eq. (A1) changes the trial in a substantial fraction of parameter sets, and at half filling the state chosen that way carries the larger staggered moment in every case where the two differ. We therefore report results obtained with Eq. (A1) throughout.

The identity $E_{\text{HF}} = \sum_p \varepsilon_p - U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle$ holds only at exact self consistency and should not be substituted for Eq. (A1). A mean-field iteration that mixes densities between steps does not reach that condition for loosely converged solutions, which are precisely the ones whose ranking is at issue.

Appendix B: Convergence in projection length

All results use a projection length $\beta = 32$ in units of $1/|t_0|$. Doubling it to $\beta = 48$ at otherwise identical settings changes Δ_{tot} by between 0.1% and 4.1% with no systematic drift in either direction, which we take as convergence.

Appendix C: Boundary conditions

At half filling with periodic boundaries the noninteracting spectrum of Eq. (1) is gapless at every L and δ examined, so the occupied set is fixed by the trial state rather than by the spectrum. Antiperiodic boundaries shift the momentum mesh off the high-symmetry points and open a gap.

The sign of $\partial\Delta_{\text{tot}}/\partial U$ agrees between the two choices at $\delta = 0.1$, where both are negative, and at $\delta = 0.5$, where both are positive, with magnitudes matching to within a few percent. At $\delta = 0.3$ the two disagree and the antiperiodic value is consistent with zero. We read $\delta \approx 0.3$ as the anisotropy at which the slope passes through zero, where its sign is most sensitive to the mesh, and we do not quote a slope there.

ACKNOWLEDGMENTS

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